

ASSIGNMENT 1 — INDUCTIVE REASONING

COMP 3010 — ORGANIZATION OF PROGRAMMING LANGUAGES

Remember: For every answer (including multiple choice) explain *how* you got your answer and *why* you think it is the correct one. This explanation will be part of your grade.

1. INDUCTION

Exercise 1 (Multiple Choice). Consider the total function

$$f(x) = 4x + 5$$

Which of the following inductive definitions are equivalent to the above f ?

- (a)

$f(x+1) = f(x) + 5$

- (b)

$f(x+1) = f(x) + 4$

- (c)

$f(0) = 5$	$f(x+1) = f(x) + 4$
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- (d)

$f(0) = 4$	$f(x+1) = f(x) + 5$
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Exercise 2. $\mathbb{N}$ stands for the set of all natural numbers, and $\mathbb{N}^*$ stands for the set of *all* finite lists of natural numbers of any length. For example, $\mathbb{N}^*$ contains each of the lists $[9, 9, 9]$, $[10, 9, 8, 7, \dots, 3, 2, 1]$ and $[5, 16, 8, 4, 2, 1]$. $\mathbb{N}^*$ is defined inductively as the smallest set such that:

- $\mathbb{N}^*$ contains the empty list $[]$, and
- given any natural number x_0 from $\mathbb{N}$ and any list $[x_1, \dots, x_n]$ already in $\mathbb{N}^*$, the set $\mathbb{N}^*$ also contains the list $[x_0, x_1, \dots, x_n]$.

Give an inductive definition for the function f , which takes a finite list of numbers xs such that $f(xs)$ is three times the sum of the list. Your definition of f should follow the inductive pattern

$$f([]) = \dots$$

$$f([x_0, x_1, \dots, x_n]) = \dots f([x_1, \dots, x_n]) \dots$$

so that it gives the same result as:

$$f([x_0, x_1, x_2, \dots, x_n]) = 3 \times (x_0 + x_1 + x_2 + \dots + x_n)$$

on any argument. In the case of the empty list $[]$, the sum of $[]$ is 0.

Hint: you might find the fact that

$$3 \times (x_0 + x_1 + \dots + x_n) = (3 \times x_0) + (3 \times x_1) + \dots + (3 \times x_n)$$

is useful for writing your inductive definition of f .

Exercise 3 (Multiple Choice). Let X be inductively defined as the smallest set of finite number lists (taken from $\mathbb{N}^*$) satisfying the following closure properties:

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- X contains the empty list $[]$.
- Given any natural number n from $\mathbb{N}$, the set X contains the one-element list $[n]$.
- Given any natural number y from $\mathbb{N}$ and any list $[x_1, \dots, x_n]$ already in X , the set X also contains the list $[y, x_1, \dots, x_n, y]$.

Which of the following lists is NOT in X ?

- (a) $[1, 2, 3, 2, 1]$
- (b) $[1, 2, 1, 2, 1]$
- (c) $[1, 2, 3, 4, 5]$
- (d) $[1, 1, 1, 1, 1]$

2. SYNTAX AND GRAMMARS

Exercise 4 (Multiple Choice). Consider the grammar G (note that ε stands for the empty string):

$$S ::= aS \mid T$$

$$T ::= bT \mid U$$

$$U ::= cU \mid \varepsilon$$

- (1) Which of the following strings is generated by the grammar G ?
 - (a) **acb**
 - (b) **acc**
 - (c) **cba**
 - (d) **baa**
- (2) Which of the following is a derivation of **bbc** in the grammar G ?
 - (a) $S \rightarrow bT \rightarrow bbT \rightarrow bbU \rightarrow bbcU \rightarrow bbc$
 - (b) $S \rightarrow T \rightarrow bT \rightarrow bTbT \rightarrow bbT \rightarrow bbU \rightarrow bbcU \rightarrow bbc$
 - (c) $S \rightarrow T \rightarrow U \rightarrow bU \rightarrow bbU \rightarrow bbU \rightarrow bbc$
 - (d) $S \rightarrow T \rightarrow bT \rightarrow bbT \rightarrow bbU \rightarrow bbcU \rightarrow bbc$

Exercise 5 (Short Answer). In your own words, describe what language this grammar generates (remember, ε stands for the empty string):

$$S ::= aSa \mid T$$

$$T ::= bT \mid \varepsilon$$

3. SYNTAX TREES

Exercise 6. Consider the following grammar for *concrete* syntax of arithmetic expressions:

$$E ::= E+T \mid E-T \mid T$$

$$T ::= T \times F \mid T/F \mid F$$

$$F ::= 1 \mid 2 \mid 3 \mid 4 \mid (E)$$

$$1 \times 2 - (3/4)$$

- (1) Write a parsing derivation starting from E (looking like $E \rightarrow E-T \rightarrow \dots$)
- (2) Draw a parse tree (remember that the parentheses in the string should be included somewhere as a node for parentheses in the tree!)

Exercise 7. Consider the following grammar for *abstract* syntax of arithmetic expressions:

$$E ::= E + E \mid E - E \mid E \times E \mid E / E \mid 1 \mid 2 \mid 3 \mid 4$$

with the usual associativity and precedence for the arithmetic operators (all operators are left-associative, $\times$ and $/$ have a higher precedence than $+$ and $-$).

- (1) Draw an abstract syntax tree for each of the following strings:
 - (a) $1 + (3/4)$
 - (b) $1 \times (2 + 3)$
 - (c) $(1 \times 2) - (3/4)$
- (2) (Multiple Choice & Short Answer) Which of the following pairs of strings with different parentheses represent the same abstract syntax tree according to the above precedence and associativity? Draw that abstract syntax tree.
 - (a) $2 \times 4 - 3$ versus $2 \times (4 - 3)$
 - (b) $1 + 2 + 3 + 4$ versus $1 + (2 + (3 + 4))$
 - (c) $2 + 3 \times 4/2$ versus $2 + ((3 \times 4)/2)$